

Controlled VLM alignment for grounded reasoning

same budget, same runner, same data caps - only the training signal changes

1 Problem

A VLM can answer correctly while using weak or wrong visual evidence.

2 Controlled study

Qwen2-VL
QLoRA

+

BLIP-2
Q-Former

answer-only, rationale, explanation-aware, and contrastive-enhanced signals

3 Evaluation

Answer quality

Rationale quality

Visual sensitivity

4 Impact

Do not trust accuracy alone.

What the study makes visible

Explanations are not grounding by default

Better rationale text can move separately from masking-based visual sensitivity.

Controls prevent over-crediting

Answer-only and generative controls show what the extra objective really adds.

Reusable single-GPU protocol

Configs, split audits, artifacts, and claims are tied together before reporting.

Impact for the study

Safer interpretation

Accuracy, rationale quality, and visual sensitivity are not merged.

Cleaner claims

Every conclusion is tied back to a control, table, or artifact.

Reusable protocol

The pipeline can be rerun by a student lab on one GPU.

Main message: do not judge grounded reasoning from final answer accuracy alone.